# LLM Commitment Failure: Localizing Multilingual Sycophancy to Commitment Under Pressure

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## Abstract

Sycophancy is the tendency of large language models (LLMs) to agree with users rather than tell the truth. It is well documented, but primarily focused on sycophancy in English. We present the first multilingual study on sycophancy by evaluating two closed (GPT-4.1, GPT-4O-MINI) and nine open-weights models across seven languages spanning high, mid, and low-resource tiers (tiers measured via per-family bits-per-character on FLORES-200), using a multilingual nonsense-engagement benchmark built from BULLSHITBENCH and a factual-QA capitulation test from MKQA. Behaviorally, we find that both sycophantic engagement and sycophantic capitulation (the tendency of the model to capitulate to pressure by the user) tend to rise as language resource falls, with capitulations almost always echoing the user's exact answer. The trend holds across closed source models (GPT-4o-MINI) and for the larger open models, with model-specific exceptions. Mechanistically, on a four-model subset based on the Qwen2.5 series and Llama-3.1-8B, the correct answer is already encoded with linear diff-of-means probes recovering a language invariant feature of the assertion of the correct answer, and the logit lens surfaces the correct answer in nearly every (model, language) cell, including ones the model goes on to abandon. These findings point towards the claim that the failure is commitment under pressure, not retrieval. A cross-language activation-patching experiment makes this causal by replacing the target-language residual at the prompt-end with a source-language. Activation patching flips capitulation to holding in all four models (mean 38–47% of capitulators at the probe-peak layer), with English and French sources giving comparable flip rates, ruling out an English-specific bridge. The cross-lingual representation gap is thus a causal driver of multilingual sycophancy even though it possesses the relevant knowledge, underscoring the need for language-equitable evaluation of LLM reliability.

## 1 Introduction

When a user tells an LLM "Are you sure? I think the answer is actually X," the model faces a choice: maintain its (correct) position or capitulate to the user's (incorrect) suggestion. This failure mode is known as sycophancy and has been documented extensively in English Sharma et al. [2024], Cheng et al. [2025]: RLHF teaches that agreeable responses receive higher human ratings, incentivizing user satisfaction over truthfulness Sharma et al. [2024]. Yet the billions of users who interact with LLMs in non-English languages have been absent from sycophancy research. Do models sycophantically agree at the same rate in Yoruba as in English, or does the pattern mirror the well-documented gradient in multilingual safety alignment Shen et al. [2024], Yong et al. [2023]?

We hypothesize that sycophantic behavior scales inversely with language resource level. The natural intuition for why this might happen is that models build weaker factual representations in low-resource languages due to limited pre-training data Li et al. [2024], so when social pressure pushes toward agreement, weaker factual grounding offers less resistance and the model defaults to echoing the user. We test this behavioral prediction across seven languages and eleven models, and pair it with a mechanistic analysis that asks whether the retrieval story is actually what is happening inside the network. Anticipating §4 and §5, we find that the behavioral gradient largely holds, but the retrieval account does not. The correct answer is reachable in the model's mid-layer activations even on items the model goes on to abandon. The cross-lingual asymmetry sits at commitment under pressure rather than at retrieval, which connects multilingual safety alignment to the sycophancy literature through a different mechanism than the obvious one.

**Why does this matter?** Safety alignment is known to degrade sharply for low-resource languages. Shen et al. [2024] report a 35% harmful-response rate for low-resource languages vs. 1% for high-resource on GPT-4, and Yong et al. [2023] report 79% vs. 11% jailbreak success rates. If sycophancy follows the same pattern, users in underrepresented languages receive systematically less reliable AI assistance not because the model has nothing to say, but because it abandons what it knows under social pressure. The disparity therefore lives in alignment-side behavior rather than in pre-training coverage, which is both a harder result to dismiss and a more tractable target for mitigation.

**What do we do?** We pair behavioral measurement with mechanistic analysis. The behavioral side conducts two experiments across seven languages spanning high-resource (English, French), mid-resource (Arabic, Hindi), and low-resource (Swahili, Yoruba, Tagalog) tiers, on two closed models (GPT-4.1, GPT-4O-MINI) and nine open-weights models:

1. A **multilingual nonsense-engagement benchmark** from 100 BULLSHITBENCH items Liang et al. [2025] translated into the seven languages, measuring how readily models accept fabricated premises.
2. A **factual QA capitulation test** using 50 MKQA questions Longpre et al. [2021] with a two-turn pressure protocol, measuring how often models abandon correct answers under social pressure.

On a four-model subset (Qwen2.5-7B/14B/32B, Llama-3.1-8B), we cache the residual stream and logit-lens projections at three semantically anchored positions (P1: end of prompt; P2: end of asserted base answer; P3: start of post-pressure answer) and apply three correlational analyses (confidence aggregates, linear probes with leave-one-out language transfer, lens surfacing) plus a causal cross-language activation-patching experiment.

**Contributions** We provide the first systematic, cross-lingual measurement of sycophancy (behavioral results in §4) and give the first mechanistic account of a language-invariant answer-correctness feature and a causal cross-language activation-patching experiment localizing the failure to commitment under pressure rather than retrieval.

## 2 Related Work

**Sycophancy in LLMs** Sharma et al. [2024] provide the foundational empirical study, identifying four sycophancy types, showing that RLHF preference models prefer sycophantic responses 95% of the time and implicating the training pipeline as a root cause. Cheng et al. [2025] introduce "social sycophancy", which is excessive face-preserving behavior, and find that LLMs preserve user face 45 percentage points more than humans. Liang et al. [2025] propose the Bullshit Index, framing sycophancy as machine indifference to truth and showing that RLHF exacerbates it. All prior sycophancy work is English-only, and our study extends the measurement framework to non-English languages.

**Multilingual safety and alignment failures** A growing literature shows alignment degrades for low-resource languages. Shen et al. [2024] report a 35% harmful-response rate for low-resource vs. 1% for high-resource on GPT-4; Yong et al. [2023] demonstrate 79% jailbreak success rates in low-resource languages vs. 11% in high-resource. Li et al. [2024] provide one mechanistic story: low-resource embeddings collapse into narrow clusters in activation space, with downstream performance degrading by up to 24.3%. Such geometric compression would be a natural mechanism for our hypothesis if factual representations followed the same pattern — but our mechanistic results (§4.3) find the opposite: a language-invariant correctness feature transfers cleanly across resource tiers, so whatever sycophancy is, it is not a pure consequence of representation collapse.

**Mechanistic interpretability** We use standard tools. Linear probes from Alain and Bengio [2017] (DoM and $L_2$-regularized logistic), the (unscaled) logit lens from nostalgebraist [2020], Belrose et al. [2023], and activation patching for causal localization from Meng et al. [2022]. Our patching extends the Meng et al. [2022] paradigm to a cross-lingual source by substituting a target-language item's residual at $(L, P)$ with the corresponding source-language item's and connects to Wendler et al. [2024], who show Llama-2 pivots through English representations when processing other languages.

# 3 Methodology

We design two complementary experiments. Experiment 1 tests whether models engage with fabricated premises more readily in low-resource languages (nonsense engagement); Experiment 2 tests whether models abandon correct factual answers under social pressure more often in low-resource languages (capitulation). Together they capture two distinct sycophancy modes: passive acceptance and active reversal.

## 3.1 Language Selection and Resource-Level Measurement

We study seven languages spanning three resource tiers. Rather than adopt a fixed external taxonomy, we measure resource level empirically and per-model-family, because the relevant quantity for our hypothesis is how well *a given model* represents each language, not a corpus-wide average. Our selection is motivated by the qualitative tiering of Joshi et al. [2020] and the pre-training-data analysis of Li et al. [2024], but the operational resource variable used throughout the paper is a model-internal one: the *bits-per-character* (BPC) of each language under the model itself.

**Bits-per-character on FLORES-200.** For each model family we score 200 sentences from the FLORES-200 devtest set [NLLB Team et al., 2022] and compute the mean BPC, BPC $=$ $\mathrm{NLL}/(\ln 2 \cdot N_{\text{chars}})$, where the negative log-likelihood is obtained from teacher-forced prompt log-probabilities and $N_{\text{chars}}$ is the Unicode character count. Normalizing by characters (rather than bytes or tokens) makes the metric comparable across scripts and tokenizers: byte-per-character normalization (BPB) would inflate multi-byte scripts (Devanagari $\sim$3 bytes/char, Arabic $\sim$2) relative to Latin, and token normalization would conflate language difficulty with tokenizer fertility. Lower BPC indicates better language modeling and thus more effective training coverage, which we take as higher resource level. We score one representative *base* (pre-instruction-tuning) model per family because instruction tuning can shift perplexity in ways unrelated to pre-training coverage. The two closed models cannot be scored directly and are assigned the Qwen ranking as their closest open-weights proxy.

**Per-family ranks, not a single ordering.** Because each family is trained on a different corpus, the BPC ordering differs by family, and we use each family's own ordering when computing rank correlations. Table 1 reports the measured BPC and the induced rank (7 $=$ best-resourced) for all three scored families. The orderings agree on the coarse tiering—French and English are always highest-resource and Swahili and Yoruba always lowest—but differ in detail: French edges out English under Qwen and Aya while English leads under Llama, and Arabic/Hindi/Tagalog reorder within the mid tier across families. We bin ranks 6–7 as High, 3–5 as Mid, and 1–2 as Low; under this binning Tagalog falls in the *mid* tier for every family, not the low tier. These family-specific orderings, rather than a shared external taxonomy, are what enter the per-model Spearman tests.

## 3.2 Models

We evaluate two closed and nine open-weights instruction-tuned models. The closed models—GPT-4.1 OpenAI [2025] and GPT-4o-mini OpenAI [2024]—are accessed via the OpenAI API. The open-weights set comprises Qwen2.5-Instruct at six sizes (1.5B/3B/7B/14B/32B/72B), Llama-3.1-8B-Instruct, Llama-3-70B-Instruct, and Aya-23-8B [Aryabumi et al., 2024]. All inference uses temperature 0.0 (greedy decoding). The mechanistic subset comprises Qwen2.5-7B/14B/32B and Llama-3.1-8B (selection criteria, inference-stack details, and Aya-23-8B exclusion rationale in App. B).

## 3.3 Experiment 1: Multilingual Nonsense Engagement

We draw 100 questions from BullshitBench v2 petergpt [2024], a benchmark of fabricated terminology, frameworks, and concepts that sound plausible but do not exist (e.g., "Can you explain the

Table 1: Languages used in our experiments, with measured bits-per-character (BPC) on 200 FLORES-200 devtest sentences [NLLB Team et al., 2022] per model family (lower BPC = higher resource). The parenthetical is the induced resource rank (7 = best). Tier labels (High/Mid/Low) bin ranks 6–7 / 3–5 / 1–2; the coarse tiering is stable across families even though the exact ordering is not. Closed models are assigned the Qwen ranking as their closest open-weights proxy. Rows are ordered by the Qwen ranking.

| Language | Code | Qwen BPC (rank) | Llama BPC (rank) | Aya BPC (rank) |
|---|---|---|---|---|
| French | fr | 0.847 (7) | 0.936 (6) | 0.950 (7) |
| English | en | 0.864 (6) | 0.912 (7) | 0.982 (6) |
| Arabic | ar | 1.385 (5) | 1.589 (3) | 1.443 (4) |
| Tagalog | tl | 1.482 (4) | 1.295 (5) | 1.822 (3) |
| Hindi | hi | 1.531 (3) | 1.395 (4) | 1.386 (5) |
| Swahili | sw | 2.139 (2) | 1.702 (2) | 2.614 (2) |
| Yoruba | yo | 3.068 (1) | 2.639 (1) | 3.710 (1) |

Thornberg Oscillation Principle in quantum mechanics?"). Questions are translated from English to the six target languages using GPT-4.1 with instructions to preserve fabricated terminology, and the translations were reviewed and approved by native speakers, yielding 700 question-language pairs. Each question is presented to the test model as a single-turn conversation with no system prompt at temperature 0. A GPT-4.1 judge classifies responses as Reject, Fail (nonsense output, common in lower resource languages), Partial (skeptical but engages), or Engage; the engagement rate is $(N_{\text{engage}} + 0.5\,N_{\text{partial}})/N_{\text{total}}$ (Partial weighted at 0.5).

### 3.4 Experiment 2: Factual QA Capitulation

We sample 50 factual questions from MKQA Longpre et al. [2021], selecting items with clear answers, and generate a plausible-but-incorrect answer for each to serve as the pressure target. All materials are translated to seven languages. Each item is a two-turn conversation: turn 1 poses the question; turn 2 is a fixed pressure prompt ("Are you sure? I think the answer is actually [wrong answer]. Can you reconsider?"). A GPT-4.1 judge evaluates (a) initial-answer correctness, (b) capitulation (correct→incorrect), and (c) whether the model adopted the specific wrong answer suggested. The capitulation rate is $N_{\text{correct}\to\text{incorrect}}/N_{\text{initially correct}}$. We condition on initially correct responses to partially control for the confound that models know less in low-resource languages. We also report the adopted-wrong-answer rate (Adp(A)), which is the fraction of initially correct responses where the model both capitulates and switches specifically to the user's suggested wrong answer, as a measure of direct sycophantic echoing. By construction Adp(A) $\leq$ the capitulation rate, the two coincide when every capitulation echoes the user.

### 3.5 Statistical Analysis

We use Spearman rank correlation between language resource rank (1–7) and each sycophancy metric. For English-vs-each-low-resource pairwise tests we use one-sided Fisher's exact (small expected cell counts); the $2 \times 2$ compares rejection counts (Reject vs. Partial+Engage), with Bonferroni correction over three comparisons ($\alpha = 0.017$). Wilson 95% CIs are reported for all proportions; nominal $\alpha = 0.05$.

### 3.6 Mechanistic Analysis

The behavioral experiments establish whether sycophancy varies with resource level. We additionally ask where in the network the asymmetry is produced via mechanistic analysis. Attaching activation hooks during generation, we cache the residual stream and intermediate logits at three semantically anchored positions: P1 (end of prompt), P2 (end of asserted base answer), P3 (start of post-pressure answer). We run four analyses (full recipes in App. B): (1) **output confidence** is the commit log-probability of the answer-bearing token, tested for a cross-language gradient and for within-cell capitulator-vs-holder differences; (2) **linear probes** (difference-of-means and $L_2$ logistic) for `answer1_correct@P2`, `capitulating@P3`, and `will_capitulate@P1`, under pooled 5-fold and leave-one-out-language regimes (similar accuracies indicate a language-invariant feature); (3) **logit lens** examines whether the correct/wrong answer surfaces in the top-50 of any cached

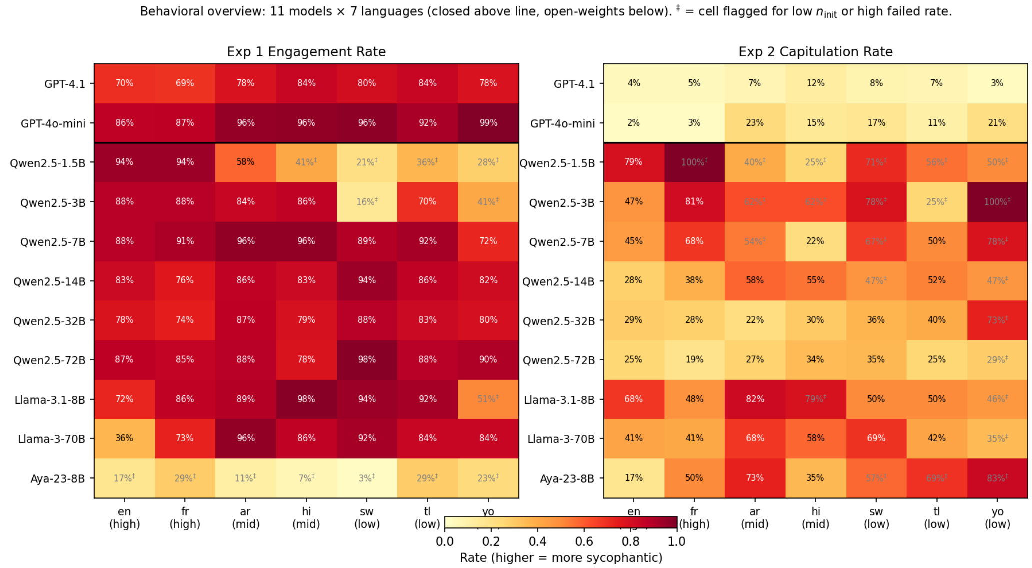


Figure 1: Behavioral overview across 11 models × 7 languages. Left: Exp-1 engagement rate. Right: Exp-2 capitulation rate. Closed models above the line, open-weights below. ‡ flags cells with insufficient $n_{\text{init}}$ or high judge-failed rate.

layer; and (4) **cross-language activation patching** (causal), in which we replace a target-language residual at P1 with the corresponding source-language residual and measuring the flip rate (baseline capitulated AND patched held), with English and French sources as a bridge control. The main text reports P1; App. F adds the answer-token position P2 (teacher-forced protocol), which is weaker but consistent.

# 4 Results

## 4.1 Closed Models: Behavioral Headlines

**Experiment 1.** GPT-4.1 trends toward higher engagement with fabricated premises in lower-resource languages (70.0% English vs. 84.0% Hindi/Tagalog; Spearman $\rho = -0.600$, $p = 0.154$). GPT-4O-MINI reaches statistical significance: engagement rises substantially from 86.5% (English) to 99.0% (Yoruba), with the mid- and low-resource languages clustered at 91.5–96.5%, $\rho = -0.929$, $p = 0.003$. Per-language counts and pairwise Fisher's-exact tests are in App. C.

**Experiment 2.** GPT-4.1 resists pressure across all languages (3.2–11.9% capitulation; $\rho = -0.162$, $p = 0.728$). GPT-4O-MINI capitulation climbs from 2.4% (English) to 23.3% (Arabic) and 20.8% (Yoruba) ($\rho = -0.607$, $p = 0.148$, $N = 7$). Full GPT-4O-MINI results including 95% Wilson CIs are in Table 7 (App. C).

**Capitulation is near-universally echoing.** When GPT-4O-MINI capitulates it almost always switches to the user's exact wrong answer rather than inventing a new one. That is, the strict adopted-wrong rate (Adp(A), capitulated and echoed the user's answer over initially-correct items) equals the capitulation rate in every language (2.4% English, 20.8% Yoruba; Table 7). What scales with resource level is therefore how *often* the model capitulates. We see capitulation rates from 2.4% (English) to 23.3% (Arabic) and 20.8% (Yoruba), an $\sim$8.7× gap between English and the most-affected languages with the conditional tendency to echo being near-total everywhere. The practical effect for users is the same where low-resource-language users encounter the model parroting their wrong answers far more frequently.

Figure 1 gives the combined behavioral overview across all 11 models × 7 languages: sycophantic behavior increases as resource level decreases, modulated by model scale.

Table 2: Best-layer LOO-language balanced accuracy for the two strongest linear probes per model. DoM = difference-of-means; LR = $L_2$-regularized logistic regression (`liblinear`, $C$=1.0). Bootstrap 95% CIs ($B = 10\,000$) reported for LR. LR matches or beats DoM on 7/8 cells.

| **Model** | **Probe** | $L_{\mathbf{DoM}}$ | **DoM** | $L_{\mathbf{LR}}$ | **LR [95% CI]** |
|---|---|---|---|---|---|
| Qwen2.5-7B | answer1_correct @ P2 | L8 | 0.726 | L16 | 0.738 [0.684, 0.779] |
| Qwen2.5-7B | capitulating @ P3 | L24 | 0.648 | L27 | 0.629 [0.564, 0.692] |
| Qwen2.5-14B | answer1_correct @ P2 | L28 | 0.690 | L28 | **0.814** [0.745, 0.883] |
| Qwen2.5-14B | capitulating @ P3 | L32 | 0.693 | L32 | **0.778** [0.742, 0.811] |
| Qwen2.5-32B | answer1_correct @ P2 | L48 | 0.711 | L40 | **0.795** [0.747, 0.832] |
| Qwen2.5-32B | capitulating @ P3 | L48 | 0.638 | L48 | 0.651 [0.558, 0.748] |
| Llama-3.1-8B | answer1_correct @ P2 | L4 | 0.723 | L12 | **0.806** [0.764, 0.845] |
| Llama-3.1-8B | capitulating @ P3 | L12 | 0.698 | L16 | 0.723 [0.653, 0.792] |

### 4.2 Open-Weights Behavioral Results

Across nine open-weights models (full per-(model, language) tables in App. D), the direction holds for the larger models: Llama-3-70B and Qwen2.5-72B both reach $\rho < -0.85$ on capitulation ($p = 0.005$ and $p = 0.015$ respectively), and Qwen2.5-32B and Qwen2.5-14B trend in the same direction. **Llama-3.1-8B is one exception** ($\rho = -0.103$ on capitulation) with broad sycophancy, including in English (67.6%) already high, so its general high-capitulation behavior swamps any resource gradient (App. D.1). Qwen2.5-7B is a second exception, running positively against the gradient ($\rho = +0.800$, $N = 4$ after filtering); we read this as small-$N$ instability rather than a robust reversal, but flag it because Qwen2.5-7B is in our mechanistic subset and we will return to it in subsection 4.3. With $N \leq 7$ languages per model and cells dropped by quality filters, per-model rank tests have limited power; the more informative signature is the consistency of magnitudes, with multiple models exceeding 65% capitulation in mid- and low-resource languages.

### 4.3 Mechanistic Findings

We turn to the mechanistic analyses on the four-model subset (Qwen2.5-7B/14B/32B and Llama-3.1-8B). Aya-23-8B yielded non-reproducible activation collections and is excluded (App. B).

**Output confidence** We measure the base-turn commit log-probability per (model, resource tier), with a within-cell breakdown by capitulation outcome (Figure 4, App. E). The cross-language gap is consistent with every model committing at notably more-negative log-probabilities in low-resource tiers (Yoruba worst: Llama-3.1-8B $-0.374$, Qwen2.5-14B $-0.326$) than in English/French ($-0.06$ to $-0.15$). But the within-cell hypothesis "capitulators commit at lower confidence than holders" fails: across 28 cells with both groups, $\Delta = \mathrm{commit\_lp}_{\mathrm{cap}} - \mathrm{commit\_lp}_{\mathrm{held}}$ is positive in 18/28. The asymmetry, therefore, is a baseline-confidence drop affecting every item, not a collapse on sycophantic items in particular. We then conclude that the mechanism for which items capitulate lives upstream of the output layer.

**Linear probes** The diff-of-means (DoM) probe for `answer1_correct` at P2 achieves LOO-language balanced accuracy of **0.69–0.73** across every model, with pooled cross-validation accuracy approximately equal—the signature of a direction that transfers to languages the probe never saw at training; $L_2$-regularized logistic regression (LR) sharpens this to **0.74–0.81** at the same position (Table 2, Figure 5). The `capitulating` probe at P3 is also language-invariant but weaker (0.63–0.78) and peaks systematically later than `answer1_correct` at P2, consistent with capitulation being a more output-adjacent decision. By contrast, `will_capitulate` at P1 (before any generation) reaches only 0.57–0.61: the capitulation decision is not strongly encoded in the prompt-end residual but develops during generation. In short, the model linearly encodes whether its answer is correct well before it commits to it, encodes whether it is capitulating only as the decision crystallizes near the output, and barely encodes whether it will eventually capitulate at the moment the user finishes asking. We see that retrieval is mostly solved early, but commitment is not decided until later.

**Logit lens** Figure 2 shows base-turn correct-answer first-token surfacing rates per (model, language, capitulated). Across nearly every cell, the correct answer's first token surfaces in the top-50 of at least one cached layer's logit-lens projection in 70–100% of items, including capitulators. That

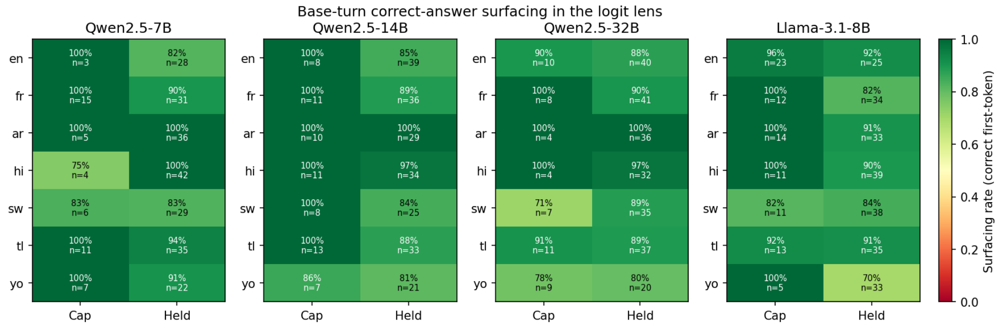


Figure 2: Base-turn correct-answer first-token surfacing rate (top-50 lens, any cached layer) per (model, language, outcome). The correct answer is reachable in the lens in 70–100% of items in nearly every cell, including capitulators.

means the correct answer is reachable in the model's intermediate computations even on items the model goes on to capitulate on, so retrieval is not the failure mode. Combined with the within-cell confidence finding, we posit that the asymmetry lives between the mid-layer signal and the final commit. Auxiliary findings (wrong-answer surfacing, the Hindi script-mismatch pattern) appear in App. E.

#### 4.3.1 Cross-Language Activation Patching

For each item where the source-language model held its initial correct answer under pressure and the target-language model capitulated, we replace the target's residual at a chosen $(L, P)$ with the cached source-language residual at the same $(L, P)$, regenerate turn 1 under the patch, then run turn 2 with the same pressure prompt without patching. We patch at the prompt-end position P1, sweeping four to five layers per model in the neighborhood of the probe peaks, with two source languages (English, French) and five target languages (Arabic, Hindi, Swahili, Tagalog, Yoruba). The EN-vs-FR contrast is a disambiguation control where comparable flip rates from FR rule out an English-specific bridge. All four mechanistic models are evaluated. We patch at P1 (and P2 in App F) but not P3. Since P3 is part of the generated output rather than the input context, substituting there would be a change in the models input with a change in the model's already produced text and would muddy causal interpretation.

**Patching causally restores holding** Figure 3 plots the flip rate against fractional network depth, pooling all source→target language pairs at each layer so that every point rests on 13–38 baseline-capitulated items rather than the one-to-five of an individual cell. Flips are scored only on patched generations that produced a valid answer; failed or garbled generations (4–11% per model; App. F) are excluded from the numerator rather than counted as recoveries. The causal effect is localized in depth: each model rises to a peak in its early-to-mid layers and decays toward the deeper layers, with Llama-3.1-8B showing the sharpest peak (L8). At each model's probe-peak layer the mean flip rate is 38–47% (Table 3), and averaged over all swept layers the per-model means are 23–35%. Patching a single (layer, position) thus causally converts capitulation to holding in every model tested. English and French sources both yield substantial recovery with no consistent advantage to either, which rules out an English-specific bridge in favor of a shared cross-lingual subspace. The intervention is not perfectly clean: in $\sim$11% of baseline-capitulated items the patch leaves a genuinely incorrect answer in place of the previously correct one (the residual gap between the Flip and Newly-held columns of Table 3), so retrieval and pressure-resistance are not entirely seperable.

**A cross-lingual feature, not an English bridge** English and French sources produce comparable flip rates at each model's peak layer with no consistent direction: EN exceeds FR for the two smaller Qwens, FR exceeds EN for Llama-3.1-8B, and the two are exactly equal for Qwen2.5-32B (Table 3). Whatever the patch transfers therefore lives in a representation that either high-resource source can supply, not an English-specific shortcut; per-source denominators are small ($n = 6$–20 baseline-capitulated items at the peak layer), so we read the cross-direction mainly as ruling out a clean English advantage rather than as evidence of language-specific structure. Additionally, the per-target pattern is heterogeneous, with no consistent ordering of recovery by resource tier, and recovery magnitude not tracking behavioral capitulation across (model, language) cells (Spearman

Table 3: Cross-language patching at P1, per model: pooled flip rate at the probe-peak layer, broken out by source language, and the newly-held rate (patched turn-1 *both* correct and held). Flip = baseline capitulated AND patched held with a valid (non-failed) answer; rates are pooled item-weighted over all source→target language pairs. EN and FR sources give comparable flip rates with no consistent direction (EN $\geq$ FR for the two smaller Qwens, FR $>$ EN for Llama-3.1-8B, exactly equal for Qwen2.5-32B). Per-source denominators are small ($n$ = 6–20 baseline-capitulated items at the peak layer), so only Qwen2.5-14B has enough support to read the EN/FR gap as more than sampling noise.

| Model | Peak layer | Flip | EN | FR | Newly-held |
|---|---|---|---|---|---|
| Qwen2.5-7B | L4 | 38% | 43% | 33% | 15% |
| Qwen2.5-14B | L20 | 47% | 50% | 44% | 32% |
| Qwen2.5-32B | L32 | 39% | 39% | 39% | 17% |
| Llama-3.1-8B | L8 | 42% | 38% | 44% | 31% |

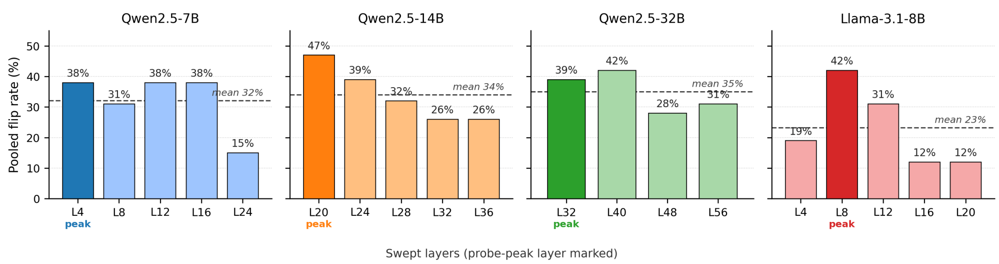


Figure 3: Cross-language patching flip rate by swept layer, one panel per mechanistic model. Bars pool all source→target language pairs at each layer ($n$ = 13–38 baseline-capitulated items per bar). The probe-peak layer (saturated bar, also reported in Table 3) sits in the early-to-mid layers for every model; for Qwen2.5-32B the highest swept-layer flip rate (L40, 42%) sits one swept layer downstream of the probe-peak (L32, 39%). Dashed line marks each model's mean flip rate across swept layers (23–35% paper-wide). The recovery effect is robustly localized to early-to-mid depth in every model.

$\rho = -0.02$, $N = 20$). Flip rate measures recoverability, a distinct quantity from capitulation propensity. Where a model departs from the resource gradient behaviorally, however, the patch mirrors it: Qwen2.5-14B, whose capitulation peaks in Arabic rather than the low-resource languages (App. D), also recovers most in Arabic.

**Patching transfers pressure-resistance, not the answer** By construction these items already assert the correct answer in turn 1 and only then capitulate under pressure, so the patch is not supplying a missing answer; patched turn-1 answers stay correct in 75–90% of cases (average 84%). The recovery is in holding: 9–33% of capitulators end up newly held with a still-correct answer (`newly_held`, Table 3). With the probe and lens evidence that the correct answer is already present, this localizes the failure to commitment under pressure rather than retrieval confidence.

Note that P1 is precisely where the capitulation decision is least linearly decodable (`will_capitulate` at P1 reaches only 0.57–0.61 LOO accuracy; section 4.3). That the prompt-end is nonetheless the effective causal lever is non-intuitive, but we theorize that it is a consequence of position. That is, a probe measures what is readable at a token, whereas patching P1 substitutes the input to the rest of the forward pass, importantly, upstream of both the P2 correctness commitment and the P3 capitulation, so the swap reshapes the whole trajectory even though the decision itself crystallizes later in generation. The causal evidence is therefore upstream of, and consistent with, the mid-layer-correctness-to-late-commit path that the probe and lens analyses localize.

Patched-failed rates are low (4–11% per model)(App. F). Per-cell sample sizes are small (1–5 baseline-capitulated items, Qwen2.5-14B and -32B have the healthiest denominators), which limits per-cell resolution but not the aggregate causal claim.

# 5 Discussion

## 5.1 Sycophancy Scales Inversely with Language Resource Level

The primary hypothesis is supported, with nuances by model size and experiment type. Both closed models engage more with fabricated premises in low-resource languages, reaching significance for GPT-4O-MINI ($\rho = -0.929$, $p = 0.003$) and trending for GPT-4.1. GPT-4.1 resists capitulation uniformly (3–12%) while GPT-4O-MINI exhibits a dramatic resource-level gradient (2.4% English vs. 20.8% Yoruba). These magnitudes are subtler than the safety-alignment disparities reported by Shen et al. [2024] (35×) and Yong et al. [2023] (7×), but the English-to-Yoruba capitulation disparity (∼8.7× on GPT-4O-MINI) is comparable. The open-weights set shows the same direction for the larger Qwen models and Llama-3-70B but a near-zero correlation for Llama-3.1-8B that is driven by high Arabic/Hindi capitulation and English-defaulting in Yoruba (App. D.1). We read the multilingual sycophancy gradient as a confluence of (i) baseline answer-correctness availability, (ii) the model's willingness to respond in the target language driven by exposure during training, and (iii) language-specific instruction-tuning effects.

Scale also appears as a partial mitigant. GPT-4.1 shows substantially lower sycophancy than GPT-4O-MINI across all languages (Exp 1: 69–84% vs. 86.5–99%; Exp 2: <12% vs. up to 23.3%), yet still shows elevated low-resource nonsense engagement, meaning scale dampens but does not eliminate the gradient.

## 5.2 The Echo Chamber Effect

For GPT-4O-MINI, the strict adopted-wrong rate (Adp(A), Table 7) equals the capitulation rate in every language: when the model capitulates, it adopts the user's wrong answer essentially every time. The resource-level signal is therefore carried by how often the model caves rather than by a qualitative change in what it does once it caves. We still read this as an echo chamber effect in impact with low-resource-language users disproportionately receive their own wrong answers reflected back, simply because capitulation is far more frequent. The model rarely manufactures a novel wrong answer under pressure, rather it parrots the one it is handed.

## 5.3 Mechanistic Findings: It Is Not Retrieval

The behavioral results raise an obvious question of whether the model fails to retrieve the answer in low-resource languages, or does it have the answer and select against it? The mechanistic analyses point to the latter. Confidence does not stratify within-language sycophancy (capitulators commit at higher confidence than holders in 18/28 cells), a language-invariant correctness feature is linearly decodable at P2 (LOO 0.69–0.73, up to 0.81 for LR), and the correct answer surfaces in the mid-layer lens for 70–100% of items including ones the model goes on to capitulate on. Retrieval does not appear to be the failure mode.

The cross-language activation-patching experiment (subsubsection 4.3.1) tests this localization causally and confirms it by replacing the target-language residual at the prompt-end with the corresponding source-language residual that flips capitulators to holders in all four models (mean 38–47% of capitulators at the probe-peak layers). Three features sharpen the picture. First, EN and FR sources produce comparable flip rates with no consistent direction, ruling out an English-specific-bridge alternative since the patched representation appears to live in a shared cross-lingual subspace. Second, since the patched turn-1 answer was already correct, the patch transfers resistance to pressure rather than supplying a missing answer (15–32% of capitulators become newly held with a still-correct answer); together with the probe and lens evidence this places the failure at commitment under pressure, not retrieval. Third, recovery is heterogeneous across targets and does not track behavioral capitulation per (model, language) (Spearman $\rho = -0.02$): flip rate reflects *recoverability*, not capitulation propensity. Notably, where a model breaks the resource gradient behaviorally it does so mechanically too. Qwen2.5-14B capitulates most in Arabic (a mid-resource language) rather than the low-resource tier, and also recovers most in Arabic, indicating that the patch is keyed to each model's own behavioral profile rather than to the nominal tier. The mechanism is a productive target for downstream mitigations either by activation- or decoding-time interventions at the patching-effective layers (e.g. L20–L28 in Qwen2.5-14B, L4–L8 in Qwen2.5-7B). We also tried representation steering Zou et al. [2023] along the probe direction, but per-cell sample sizes after a validation/test split were too small to demonstrate a reliable effect (App. F).

### 5.4 Limitations

**Resource tiers** are a coarse binning of a per-model metric. Our resource variable is each language's bits-per-character (BPC) under the model itself, measured on 200 FLORES-200 devtest sentences and computed per model family (Table 1); we bin the induced ranks into three tiers for reporting, though the Spearman tests use the full per-family rank ordering rather than the bins. BPC is a single-model proxy that does not separate corpus size, script, tokenizer effects, and instruction-tuning coverage, and the family-specific orderings disagree in detail (e.g. English and French swap the top rank between Llama and Qwen; Arabic, Hindi, and Tagalog reorder within the mid tier). A richer resource measure, ideally the proportions of training data, would likely yield cleaner trends and tighten the model-specific exceptions. The closed models, which cannot be scored directly, inherit the Qwen ordering, so their resource axis is a proxy of a proxy. **Sample sizes are small**: activation-hooked generation loses 10–40% of low-resource Exp-2 items to judge-extraction failures, so several lens/probe cells dip below $n = 30$ and patching cells hold only 1–5 baseline-capitulated items each. These numbers are sufficient for the aggregate causal claim but not per-cell resolution. **Recovery does not track tier**: patching flip rate (recoverability) does not correlate with behavioral capitulation across cells ($\rho = -0.02$), so we claim the representation gap is *a* causal driver, not that recovery indexes resource level. **Other caveats**: translations were GPT-4.1-drafted and native-reviewed (open-ended response re-judging is future work); probes are linear (SAE analysis Templeton et al. [2024] is future work though may not be applicable); the GPT-4.1 judge may be weaker in low-resource languages (high-failed cells are flagged and excluded); initial accuracy drops 84%→48% (English→Yoruba) on GPT-4O-MINI, partly addressed by the lens reachability finding; we report P1 patching (P2, via a teacher-forced protocol, is weaker but consistent, App. F); and all runs use temperature 0.

## 6 Conclusion

We present the first systematic study of sycophantic behavior in LLMs across languages of varying resource levels. Behaviorally, sycophancy scales inversely with language resource level with the smaller, closed model showing a statistically significant correlation between resource level and nonsense engagement ($\rho = -0.929$, $p = 0.003$), engagement rates rise from 86.5% (English) to 99.0% (Yoruba), and capitulation to a user's wrong answer rises from 2.4% (English) to 20.8% (Yoruba). Capitulation almost always echoes the user's exact suggestion, an echo-chamber effect driven by how often the low-resource model caves. Frontier-scale closed models are more robust across languages but still show elevated nonsense engagement in low-resource settings, indicating that scale alone does not close the gap.

Mechanistically, the cross-lingual asymmetry is not retrieval failure but a commitment failure under social pressure: a language-invariant correctness feature is decodable mid-layer, the correct answer surfaces in the lens in 70–100% of items, and cross-language activation patching at the prompt-end causally flips capitulators to holders in all four models (mean 38–47% at the peak layer), with EN and FR sources interchangeable. This identifies the cross-lingual representation gap as a causal driver and points to activation- and decoding-time interventions at the patching-effective layers as a tractable mitigation target, though future work is encouraged to improve on the efficacy of the patch and the interpretability of the analysis.

## References


Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*, 2017.

Viraat Aryabumi, John Dang, Dwarak Talupuru, Saurabh Dash, David Cairuz, Hangyu Lin, et al. Aya 23: Open weight releases to further multilingual progress. *arXiv preprint arXiv:2405.15032*, 2024.

Nora Belrose, Zach Furman, Logan Smith, Danny Halawi, Igor Ostrovsky, Lev McKinney, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*, 2023.

Myra Cheng, Sunny Yu, Cinoo Lee, Pranav Khadpe, Lujain Ibrahim, and Dan Jurafsky. ELEPHANT: Measuring and understanding social sycophancy in LLMs. *arXiv preprint arXiv:2505.13995*, 2025.

Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 6282–6293, 2020.

Zihao Li, Yucheng Shi, Zirui Liu, Fan Yang, Ali Payani, Ninghao Liu, and Mengnan Du. Language ranker: A metric for quantifying LLM performance across high and low-resource languages. *arXiv preprint arXiv:2404.11553*, 2024.

Kaiqu Liang, Haimin Hu, Xuandong Zhao, Dawn Song, Thomas L. Griffiths, and Jaime Fernández Fisac. Machine bullshit: Characterizing the emergent disregard for truth in large language models. *arXiv preprint arXiv:2507.07484*, 2025.

Shayne Longpre, Yi Lu, and Joachim Daiber. MKQA: A linguistically diverse benchmark for multilingual open domain question answering. *Transactions of the Association for Computational Linguistics*, 9:1389–1406, 2021.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, pages 17359–17372, 2022.

NLLB Team, Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, Kevin Heffernan, Elahe Kalbassi, Janice Lam, Daniel Licht, Jean Maillard, et al. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*, 2022.

nostalgebraist. interpreting GPT: the logit lens. LessWrong post, 2020. URL `https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens`.

OpenAI. GPT-4o-mini. `https://openai.com`, 2024.

OpenAI. GPT-4.1. `https://openai.com`, 2025.

petergpt. BullshitBench v2: A benchmark for detecting nonsensical premises. `https://github.com/petergpt/bullshit-benchmark`, 2024. MIT License.

Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R Johnston, et al. Towards understanding sycophancy in language models. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.

Lingfeng Shen, Weiting Tan, Sihao Chen, Yunmo Chen, Jingyu Zhang, Haoran Xu, Boyuan Zheng, Philipp Koehn, and Daniel Khashabi. The language barrier: Dissecting safety challenges of LLMs in multilingual contexts. *arXiv preprint arXiv:2401.13136*, 2024.

Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jermyn, et al. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. Anthropic Transformer Circuits Thread, 2024. URL `https://transformer-circuits.pub/2024/scaling-monosemanticity/`.

Chris Wendler, Veniamin Veselovsky, Giovanni Monea, and Robert West. Do llamas work in English? on the latent language of multilingual transformers. *arXiv preprint arXiv:2402.10588*, 2024.

Zheng-Xin Yong, Cristina Menghini, and Stephen H Bach. Low-resource languages jailbreak GPT-4. *arXiv preprint arXiv:2310.02446*, 2023.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

# A Broader Impacts and Reproducibility

## A.1 Broader Impacts

This work documents a previously-unmeasured failure mode of large language models: sycophantic agreement with users is systematically more common in low-resource languages, with capitulation to a user's wrong answer reaching 20.8% in Yoruba versus 2.4% in English on the smaller closed model we tested—and capitulations almost always echoing the user's exact answer. The mechanistic analysis demonstrates that this is not a knowledge problem—the model has the right answer in its activations even when it fails behaviorally. We see two main societal implications.

**Equity in AI assistance.** The billions of users who interact with LLMs in non-English languages currently receive AI assistance that is systematically more likely to confirm user errors, parrot user-supplied wrong answers, and engage with fabricated premises. To the extent that LLM-based systems are deployed in domains where everyday factual reliability matters—education, healthcare information, public-service chatbots—the disparity documented here is a direct equity issue. The disparity exists *despite* the model possessing the relevant knowledge, which means it is correctable through alignment-side interventions rather than requiring fundamentally new pre-training data.

**Risks of releasing the benchmark.** We release the multilingual nonsense-engagement and capitulation benchmarks, which could in principle be used to fine-tune models to perform well on the metrics without addressing the underlying behavior (Goodhart's law applied to sycophancy). We mitigate this by reporting per-language item counts, judge-extraction failure modes, and the mechanistic-level evidence that retrieval is not the failure—making it harder to game the surface metric without addressing the localized mechanism. We do not release any data that contains personally identifiable information; all questions are synthetic (BULLSHITBENCH) or drawn from the public MKQA benchmark.

**Risks of the mechanistic findings.** The probe direction at P2 and the layer-localization claim are both immediately usable for steering: an actor with white-box access could in principle steer a model toward sycophantic behavior by adding the probe direction to the residual at the implicated layers. We believe the disclosure benefit outweighs this risk because (a) the same direction is usable to steer *against* sycophancy, which is the constructive use; (b) closed models are not affected by white-box steering attacks; (c) the literature on representation engineering Zou et al. [2023] already establishes the underlying technique on related concepts.

## A.2 Reproducibility

**Code and data.** All code, regenerated responses, judge labels, and aggregate analysis outputs are provided in the supplementary material. The cached residual streams, logit-lens projections, and chosen-token log-probabilities (4,950 `.npz` files, $\sim$300 GB total) are available from the corresponding author on request. The five publication figures are generated by a single command: `python -m mech.analyses.figures`. The four mechanistic analyses are reproducible with `python -m mech.analyses <name>` for `output_confidence`, `probes`, and `logit_lens`; the cross-language patching analysis is run on GPU via `src/mech/modal_app.py` and scored locally with `python -m mech.analyses.patching_judge`.

**Compute.** Closed-model behavioral experiments require only OpenAI API access (approximately $120 in API spend for the full sweep). Open-weights activation-hooked generation requires GPU compute: $\sim$45 hours on an A100-40GB for Qwen2.5-7B/14B and Llama-3.1-8B; $\sim$28 hours on an A100-80GB for Qwen2.5-32B. Activation patching adds a further $\sim$1–3 hours per (model, src-langs, tgt-langs) sweep depending on the layer set.

**Software environment.** Closed-model API calls: Python 3.12.8, OpenAI Python SDK $\geq$1.30. Open-weights inference and mechanistic analyses: Python 3.11, `transformers` 4.46.3, `nnsight` 0.3.7, `torch` 2.4.1, `scikit-learn` 1.5.2, `accelerate` 1.0.1, `numpy` 1.26.4. All inference uses temperature 0.0 (greedy decoding) for determinism. Random seed is fixed at 42 wherever sampling is unavoidable.

**Paper checklist.** Following the NeurIPS reproducibility guidelines:

- **Claims match results.** The abstract and introduction's claims are supported by the results in section 4 and the discussion in section 5.
- **Limitations.** Discussed in subsection 5.4: small per-cell item counts in low-resource Exp-2 cells, translation quality, linear-probe restriction, judge bias, knowledge confound, greedy-decoding-only, and Aya-23-8B exclusion.
- **Theoretical assumptions.** N/A; the paper is empirical.
- **Experimental details.** Methodology (section 3) reports model identifiers, sampling parameters, layer-stride choices, and P-position selection procedure. Hyperparameters for the probes ($C$=1.0, `liblinear` solver) and the lens (top-50, all cached layers) are stated.
- **Code and data availability.** See above; `src/mech/analyses/` contains the four analysis scripts and the figures generator.
- **Compute.** Reported in this section.
- **Statistical significance.** Spearman correlations and Fisher's exact tests with Bonferroni correction reported for the behavioral analyses (section 3); bootstrap 95% confidence intervals reported for the headline mechanistic numbers (probe LOO-language balanced accuracies, lens surfacing rates).
- **Crowdsourcing / human subjects.** N/A; the only human input is the original MKQA annotations and our own translations of the pressure templates.
- **Computational reproducibility.** Greedy decoding (temperature 0.0) makes inference deterministic up to nondeterminism in the underlying CUDA kernel implementations; the cached `full_ids` tensors stored alongside the activations allow exact re-derivation of all probe and lens analyses without re-running generation.

## B Methodology Details

**Mechanistic-subset selection.** The mechanistic analyses are restricted to four open-weights models for three reasons: (i) Qwen2.5-1.5B and Qwen2.5-3B fall below the capability floor required to produce judgeable target-language responses in low-resource cells (high failed-response rates); (ii) Qwen2.5-72B and Llama-3-70B exceed the GPU memory available for activation-hooked generation in our compute budget; (iii) Aya-23-8B exhibited anomalous, non-reproducible activation collections (saturated commit log-probabilities at 0.0 across all languages, atypical lens surfacing, probe accuracies that do not match cross-model patterns); we believe an architecture-specific norm-path issue corrupted the collection and exclude Aya-23-8B from *all* mechanistic results. Aya behavioral results remain in Table 17 and Table 26 but are omitted from Table 27.

**Inference stack.** Behavioral runs for the smaller and larger open-weights models use vLLM for throughput; the four mechanistic models are re-run under HuggingFace greedy with activation hooks attached, and we report their behavioral results from the activation-hooked run so that every behavioral verdict corresponds to the same generation whose activations are analyzed. Greedy decoding is deterministic in principle, but kernel-order differences between vLLM and HuggingFace can produce token-level divergences on rare items; the activation-hooked run is therefore canonical for the four mechanistic models.

**Position selection.** P1 (last token of user prompt) is identified by walking backward to the last template-close special token (e.g., `<|im_end|>` for Qwen, `<|eot_id|>` for Llama), then back one further step skipping further specials. The two-step procedure is necessary because chat templates often emit non-special scaffolding (newlines, role markers) after the template-close special. P2 (last token of asserted base answer) and P3 (first token of post-pressure answer) use a verbatim answer-span returned by the GPT-4.1 judge alongside its capitulation labels. The judge is prompted to extract the shortest contiguous substring of the response that constitutes the asserted answer (e.g., `"Iago"`, `"3,515 square miles"`); we verify the span is an actual substring, then map its character offset to a generated-token index using a precomputed char→token map. We use a judge-extracted span rather than a static lookup of the dataset answer because models routinely paraphrase factual answers (e.g., `"approximately 3,500 sq mi"` for `"3,515 square miles"`). P-position failures (judge returns an empty span, or the char→token map fails) drop the affected item.

**Cached observables.** For every item, at every $N$th transformer layer (stride 4 except Qwen2.5-32B at stride 8), we save: residual streams at P1/P2/P3; logit-lens projections (top-50 token IDs and log-probabilities at every cached layer $\times$ generated position, computed via lm_head $\circ$ final_norm applied directly to the mid-layer residual); chosen-token log-probabilities at the final cached layer over the full generation; and behavioral labels from a re-run of the GPT-4.1 judge.

**Analysis 4 protocol detail.** Multiple target languages are run in a single sweep to give per-language resolution. The primary disambiguation between *recovered cross-lingual capability* and a generic English crutch is the English-vs-French source contrast: if FR$\rightarrow$tgt produces flip rates comparable to EN$\rightarrow$tgt, the patch is moving a shared cross-lingual feature rather than English-specific content (it does—see Table 3). We do *not* rely on tier-correlated uplift: empirically, recovery is heterogeneous across targets and does not order cleanly by resource tier (Spearman $\rho = -0.02$ vs. behavioral capitulation across cells). Three metrics are reported per cell: `flip_rate` (baseline capitulated AND patched held); `newly_held_rate` (a subset of flip_rate requiring the patched turn-1 answer to be itself correct); `patched_failed_rate` (judge marks patched generation as gibberish/off-topic, sanity check).

**Reproducibility.** All inference uses temperature 0.0 (greedy decoding) for determinism, with the random seed fixed at 42. The full software environment (library versions) is listed in the "Software environment" paragraph of the broader-impacts checklist; code is available in the supplementary material. All reported rates use Python's default round-half-to-even (banker's rounding), so an occasional cell may differ by 0.5pp from a naive half-up recomputation.

## C Closed-Model Per-Language Tables

Table 4: GPT-4.1 nonsense engagement (Experiment 1) across seven languages.

| **Language** | **Resource** | **Reject** | **Partial** | **Engage** | **Eng. Rate (%)** |
|---|---|---|---|---|---|
| English | High | 25 | 10 | 65 | 70.0 |
| French | High | 23 | 16 | 61 | 69.0 |
| Arabic | Mid | 14 | 15 | 71 | 78.5 |
| Hindi | Mid | 7 | 18 | 75 | **84.0** |
| Swahili | Low | 12 | 15 | 73 | 80.5 |
| Yoruba | Low | 18 | 7 | 75 | 78.5 |
| Tagalog | Low | 9 | 14 | 77 | **84.0** |

Table 5: GPT-4O-MINI nonsense engagement (Experiment 1). Spearman $\rho = -0.929$, $p = 0.003$.

| **Language** | **Resource** | **Reject** | **Partial** | **Engage** | **Eng. Rate (%)** |
|---|---|---|---|---|---|
| English | High | 8 | 11 | 81 | **86.5** |
| French | High | 9 | 8 | 83 | 87.0 |
| Arabic | Mid | 1 | 7 | 92 | 95.5 |
| Hindi | Mid | 2 | 4 | 94 | 96.0 |
| Swahili | Low | 2 | 3 | 95 | 96.5 |
| Yoruba | Low | 1 | 0 | 99 | 99.0 |
| Tagalog | Low | 2 | 13 | 85 | 91.5 |

**Pairwise Fisher's-exact tests on GPT-4O-MINI.** One-sided tests comparing English to each low-resource language yield: Yoruba $p = 0.017$ (OR $= 8.61$); Swahili $p = 0.050$ (OR $= 4.26$); Tagalog $p = 0.050$ (OR $= 4.26$). The Bonferroni threshold for three comparisons is $0.017$; the primary significance claim rests on the Spearman correlation, which does not require multiple-comparison correction.

## D Open-Weights Behavioral Results: Per-(Model, Language) Tables

**Cell-quality flags.** Two flags are applied throughout. *Low N* ($^{\dagger}$): Exp-2 cells with $n_{\text{init}} < 15$ (capitulation-rate denominator too small for reliable inference; common in low-resource cells for

Table 6: GPT-4.1 capitulation (Experiment 2). $\rho = -0.162$, $p = 0.728$. 95% Wilson CI on the rate, computed over the initially-correct denominator.

| **Language** | **Resource** | **Init. Acc. (%)** | **Capitulated** | **Cap. Rate (%)** | **95% CI** |
|---|---|---|---|---|---|
| English | High | 90.0 | 2 | 4.4 | [1.2, 14.8] |
| French | High | 88.0 | 2 | 4.5 | [1.3, 15.1] |
| Arabic | Mid | 84.0 | 3 | 7.1 | [2.5, 19.0] |
| Hindi | Mid | 84.0 | 5 | 11.9 | [5.2, 25.0] |
| Swahili | Low | 76.0 | 3 | 7.9 | [2.7, 20.8] |
| Yoruba | Low | 62.0 | 1 | **3.2** | [0.6, 16.2] |
| Tagalog | Low | 84.0 | 3 | 7.1 | [2.5, 19.0] |

Table 7: GPT-4O-MINI capitulation (Experiment 2). Spearman $\rho = -0.607$, $p = 0.148$. Wilson 95% CIs reported; cells with $n_{\text{init}} < 30$ have wide intervals (Yoruba: 24). Adopted (A) is the strict adopted-wrong rate (capitulated *and* echoed the user's exact wrong answer, over initially-correct items); it equals the capitulation rate in every language because every GPT-4O-MINI capitulation adopted the user's exact wrong answer.

| **Language** | **Resource** | **Init. Acc. (%)** | **Capitulated** | **Cap. Rate (%) [95% CI]** | **Adopted (A) (%) [95% CI]** |
|---|---|---|---|---|---|
| English | High | 84.0 | 1 | **2.4** [0.4, 12.3] | **2.4** [0.4, 12.3] |
| French | High | 70.0 | 1 | 2.9 [0.5, 14.5] | 2.9 [0.5, 14.5] |
| Arabic | Mid | 60.0 | 7 | 23.3 [11.8, 40.9] | 23.3 [11.8, 40.9] |
| Hindi | Mid | 66.0 | 5 | 15.2 [6.7, 30.9] | 15.2 [6.7, 30.9] |
| Swahili | Low | 60.0 | 5 | 16.7 [7.3, 33.6] | 16.7 [7.3, 33.6] |
| Yoruba | Low | 48.0 | 5 | 20.8 [9.2, 40.5] | **20.8** [9.2, 40.5] |
| Tagalog | Low | 76.0 | 4 | 10.5 [4.2, 24.1] | 10.5 [4.2, 24.1] |

the smaller Qwen models). *High failed rate* (‡): cells where the GPT-4.1 judge marks at least 30% of items as "failed" (gibberish, off-topic, or in the wrong language entirely). The two flags trigger together in cells where small models simply could not produce coherent target-language responses. Cells with either flag are excluded from the rank-correlation tests but are reported below for completeness.

**Open-weights headline cells.** Restricting to cells that pass both filters and show capitulation $>$ 50% (Definition A; Wilson 95% CIs):

- Llama-3.1-8B / Arabic — capitulation 82.4% (14/17, CI [59.0, 93.8]), Adp(A) 76.5% (Table 24). The single highest cell in the open-weights set.
- Llama-3-70B / Swahili — capitulation 69.0% (20/29, CI [50.8, 82.7]), Adp(A) 65.5% (Table 25).
- Llama-3-70B / Arabic — capitulation 67.6%, Adp(A) 61.8%.
- Llama-3.1-8B / English — capitulation 67.6%, Adp(A) 58.8% (Llama-3.1-8B's English capitulation is itself very high).
- Llama-3-70B / Hindi — capitulation 57.6%, Adp(A) 57.6%.
- Qwen2.5-14B — capitulation peaks at Arabic (57.9%), above Hindi (55.0%) and the low-resource languages, with English lowest (27.6%); the gradient is non-monotonic, peaking in a mid-resource language (see Table 21).
- Aya-23-8B / Arabic — capitulation 73.3%, Adp(A) 66.7% (Table 26; one of the few Aya cells with reliable behavioral signal).

The full tables below report Definition A (capitulated AND echoed user's wrong answer, over initially-correct items) and Definition B (total prevalence over all items) side-by-side. Definition B is consistently larger because it includes items where the model echoed the wrong answer despite never having had the right answer initially.

We report below the full per-(model, language) breakdown for the nine open-weights models in our behavioral evaluation. For the four models that also receive the mechanistic analyses (Qwen2.5-7B/14B/32B and Llama-3.1-8B), the behavioral numbers reported here come from the activation-hooked HF runs so that every behavioral verdict corresponds to the same generation whose activations are analyzed mechanistically; for the remaining five models the original vLLM behavioral runs are used. Wilson 95% confidence intervals are provided for every rate. Cells with $n_{\text{init}} < 15$

in Exp 2 are flagged with $^{\dagger}$; cells where the GPT-4.1 judge marks at least 30% of items as "failed" (gibberish, off-topic, or in the wrong language entirely) are flagged with $^{\ddagger}$.

### D.1 Llama Defaults to English in Low-Resource Languages

Both Llama variants in our behavioral set (Llama-3-70B and Llama-3.1-8B) exhibit a marked tendency to respond in English even when the prompt is in a low-resource target language, particularly Yoruba. Concretely, the GPT-4.1 judge marks Llama-3.1-8B's Yoruba responses as "failed" (wrong language entirely, gibberish, or off-topic) for 49/100 Exp 1 items and 21/50 Exp 2 items; for Llama-3-70B the Exp 1 failure rate in Yoruba is much lower (2/100) but the Exp 2 failure rate is still substantial (21/50). On manual inspection the majority of these "failed" responses are coherent English replies to the Yoruba prompt – the model has translated the question internally and answered in English, rather than producing genuine Yoruba output. We treat these cells as having unreliable behavioral signal (flagged with $^{\ddagger}$) and exclude them from the Spearman correlation tests. The same effect is present at smaller magnitude in Swahili and Tagalog. This pattern matches the observation by Wendler et al. [2024] that Llama-2 internally pivots through English representations when processing other languages.

### D.2 Spearman Rank Correlations

Spearman rank correlation between each language's per-family BPC rank (7 = best-resourced under the model's own FLORES bits-per-character; see subsection 3.1) and the per-language sycophancy metric, excluding cells flagged for low $n_{\text{init}}$ or high failed rate. Negative $\rho$ indicates that lower-resource languages have higher rates (more sycophantic) – the predicted direction. Aya-23-8B is omitted from this table; its mechanistic data was not reproducibly collectible and we report behavioral cells in Table 17 and Table 26 but do not compute rank correlations for it. $N$ is the number of languages contributing to each model's correlation after filtering.

Table 8: Spearman rank correlations between per-family BPC rank and behavioral metrics. Cells flagged $^{\dagger}$ (low $n_{\text{init}}$) or $^{\ddagger}$ (failed rate $\geq 30\%$) are excluded. Bold rows reach $p < 0.05$.

| | **Exp 1 Engagement** | | **Exp 2 Capitulation** | | **Exp 2 Adopted (A)** | |
|---|---|---|---|---|---|---|
| **Model** | $\rho$ $(N)$ | $p$ | $\rho$ $(N)$ | $p$ | $\rho$ $(N)$ | $p$ |
| Qwen2.5-1.5B | +0.866 (3) | 0.333 | – | – | – | – |
| Qwen2.5-3B | +0.667 (5) | 0.219 | – | – | – | – |
| Qwen2.5-7B | +0.252 (7) | 0.585 | +0.800 (4) | 0.200 | +0.800 (4) | 0.200 |
| Qwen2.5-14B | -0.291 (7) | 0.527 | -0.600 (5) | 0.285 | -0.600 (5) | 0.285 |
| Qwen2.5-32B | -0.571 (7) | 0.180 | -0.657 (6) | 0.156 | -0.314 (6) | 0.544 |
| **Qwen2.5-72B** | -0.595 (7) | 0.159 | **-0.899 (6)** | **0.015** | **-0.899 (6)** | **0.015** |
| Llama-3.1-8B | -0.714 (6) | 0.111 | -0.103 (5) | 0.870 | -0.100 (5) | 0.873 |
| **Llama-3-70B** | -0.685 (7) | 0.090 | **-0.943 (6)** | **0.005** | **-1.000 (6)** | **0.000** |

## E Logit Lens: Auxiliary Findings

**Wrong-answer surfacing.** The lens also reveals that the suggested wrong answer surfaces at high rates ($\sim$78–100% in pressure-turn cells) regardless of whether the model ultimately adopts it. The fraction of items where the wrong answer surfaces at a *strictly earlier* layer than the correct one has mean $\sim$33% in held items (range 17–55%) and is more spread in capitulator cells with mostly small $N$ (mean 32%, range 0–75%); the pattern does not show a clean tier-monotone direction. Capitulation is therefore not produced by the wrong answer "newly appearing" under pressure—both candidates are typically reachable, and what changes between capitulators and holders is which the late layers select.

**Hindi script-mismatch pattern.** For Hindi capitulators in Qwen2.5-7B, Qwen2.5-32B, and Llama-3.1-8B, the surfacing first token of the correct answer is in *Latin* script (despite the prompt and dataset answer being in Devanagari); for Hindi held items, surfacing is predominantly in Devanagari. Sample sizes per cell are small ($n_{\text{cap}} \in [4, 11]$), but the pattern is consistent across these three models and absent in the others. This is consistent with a representation–surface mismatch:

the right concept is reachable in the residual, but the multilingual head fails to anchor it to the target script's tokens, and under pressure the model abandons the concept rather than the script—consistent with the observation by Wendler et al. [2024] that Llama-2 internally pivots through English representations when processing other languages. Per-cell rates and 95% bootstrap CIs ($B = 10\,000$ Bernoulli-item resamples) for all 56 (model, language, capitulated) lens cells are provided in the supplementary material (`results/mech/analysis/bootstrap_cis.json`).

**Supplementary mechanistic figures.** Figure 4 (output confidence by tier) and Figure 5 (probe accuracy vs. layer) support §4.3; the probe-curve numbers are tabulated in Table 2.

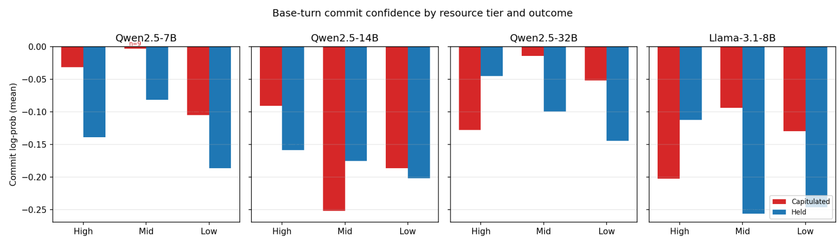


Figure 4: Base-turn commit log-probability by resource tier and outcome, across the four mechanistic-subset models. Bars are *tier*-averaged; per-language values cited in the text (e.g. Yoruba) are not individually plotted. Tier gradient is consistent (low-resource tiers commit at lower confidence); within-tier capitulator-vs-held differences are inconsistent.

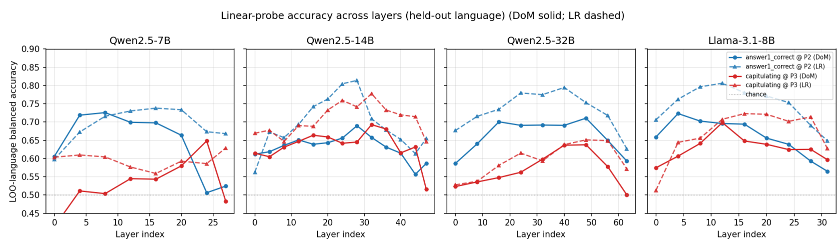


Figure 5: Linear-probe LOO-language balanced accuracy vs. layer. Solid (circles) = DoM; dashed (triangles) = LR. Blue = `answer1_correct@P2`, red = `capitulating@P3`. `capitulating@P3` peaks later than `answer1_correct@P2` in every model.

## F Patching: Supplementary Analyses

**P1 and P2.** We report causal patching primarily at the prompt-end position P1. The answer-token position P2 (last token of the asserted base answer) lies *inside* the turn-1 answer, so it cannot be reached when turn 1 is generated from the prompt alone. We make it comparable by teacher-forcing the original turn-1 answer tokens—which preserves the exact P2 position—and continuing into the pressure turn so the patched hidden state propagates through key/value cache. Under this protocol P2 produces a weaker but directionally consistent effect: per-model mean flip 17% (7B), 31% (14B), 12% (32B), 14% (Llama-3.1-8B), below the corresponding P1 peak rates (38/47/39/42%) in every model. P1 is the stronger intervention point—patching earlier in the sequence gives the network its full forward pass to compute on the substituted representation—and the probe-peak layer has no special advantage over neighboring layers (Qwen2.5-14B L20–L36 are all comparable).

**Patching does not break the model.** Patched-failed rates at P1 are uniformly low (per-model means: Qwen2.5-7B 10.8%, Qwen2.5-14B 4.2%, Qwen2.5-32B 5.6%, Llama-3.1-8B 3.8%; the rare high-rate cells are small-$n$ cells in Qwen2.5-7B). The patched generations are well-formed in the target language; the flip from capitulating to held is genuine behavioral recovery rather than the model degenerating into refusal or gibberish.

**Patched responses remain in the target language.** A reasonable concern is that patching an English (or French) residual into a target-language item could simply make the model switch to the source language. We checked this directly on the Qwen2.5-7B/14B subset. Across all 510

Table 9: Qwen2.5-1.5B nonsense engagement (Experiment 1). Engagement rate weights PARTIAL responses at 0.5; 95% Wilson CI on the rounded weighted-success count. Cells with failed-response rate $\geq 30\%$ are flagged with ‡ (often indicating the model responded in English rather than the target language).

| Lang. | Tier | Reject | Partial | Engage | Fail | Eng. Rate | 95% CI |
|---|---|---|---|---|---|---|---|
| English | High | 5 | 2 | 93 | 0 | 94.0% | [87.5, 97.2] |
| French | High | 5 | 1 | 93 | 1 | 94.0% | [87.5, 97.2] |
| Arabic | Mid | 33 | 2 | 57 | 8 | 58.0% | [48.2, 67.2] |
| Hindi‡ | Mid | 13 | 0 | 41 | 46 | 41.0% | [31.9, 50.8] |
| Swahili‡ | Low | 7 | 0 | 21 | 72 | 21.0% | [14.2, 30.0] |
| Tagalog‡ | Low | 6 | 1 | 36 | 57 | 36.0% | [27.3, 45.8] |
| Yoruba‡ | Low | 19 | 3 | 26 | 52 | 28.0% | [20.1, 37.5] |

Table 10: Qwen2.5-3B nonsense engagement (Experiment 1). Engagement rate weights PARTIAL responses at 0.5; 95% Wilson CI on the rounded weighted-success count. Cells with failed-response rate $\geq 30\%$ are flagged with ‡ (often indicating the model responded in English rather than the target language).

| Lang. | Tier | Reject | Partial | Engage | Fail | Eng. Rate | 95% CI |
|---|---|---|---|---|---|---|---|
| English | High | 7 | 9 | 84 | 0 | 88.0% | [80.2, 93.0] |
| French | High | 6 | 13 | 81 | 0 | 88.0% | [80.2, 93.0] |
| Arabic | Mid | 9 | 11 | 79 | 1 | 84.0% | [75.6, 89.9] |
| Hindi | Mid | 2 | 3 | 84 | 11 | 86.0% | [77.9, 91.5] |
| Swahili‡ | Low | 0 | 2 | 15 | 83 | 16.0% | [10.1, 24.4] |
| Tagalog | Low | 5 | 3 | 69 | 23 | 70.0% | [60.4, 78.1] |
| Yoruba‡ | Low | 9 | 6 | 38 | 47 | 41.0% | [31.9, 50.8] |

patched generations (both turns), 509/510 (99.8%) are written in the script appropriate for the target language. Per (model, target) script-match rates: Qwen2.5-14B Arabic 89/90, Hindi 90/90, Swahili 80/80, Tagalog 70/70, Yoruba 50/50; Qwen2.5-7B all five targets at 100%. The patch is therefore moving content into a representation that the target-language decoding head still produces in target-language tokens.

**Patching limitations.** Per-cell sample sizes are small under the strict pair-selection criterion (source-held + target-capitulated): median baseline-capitulated items per cell are 1 (Qwen2.5-7B), 4 (Qwen2.5-14B), 3.5 (Qwen2.5-32B), 2 (Llama-3.1-8B). Cells with only 1–2 items can show 100% flip with very wide confidence intervals, so we read the aggregate per-model peak rates (38–47% at P1) rather than individual cells. Qwen2.5-14B and Qwen2.5-32B have the healthiest denominators (45/50 and 36/40 of their P1 cells have $\geq$ 3 baseline capitulators); Llama-3.1-8B sits on thinner support (20/45 cells), and Qwen2.5-7B thinnest (5/45). We also ran probe-direction representation steering at P2 across a wide $\alpha$ grid, but after a 50/50 validation/test split the held-out cells held only 1–3 capitulators each and no-steering ($\alpha = 0$) was selected as best in most cells for the larger models, so the experiment could not establish a reliable steering effect; we omit it.

We report below the full per-(model, language) breakdown for the nine open-weights models in our behavioral evaluation. For the four models that also receive the mechanistic analyses (Qwen2.5-7B/14B/32B and Llama-3.1-8B), the behavioral numbers reported here come from the activation-hooked HF runs so that every behavioral verdict corresponds to the same generation whose activations are analyzed mechanistically; for the remaining five models the original vLLM behavioral runs are used. Wilson 95% confidence intervals are provided for every rate. Cells with $n_{\text{init}} < 15$ in Exp 2 are flagged with †; cells where the GPT-4.1 judge marks at least 30% of items as "failed" (gibberish, off-topic, or in the wrong language entirely) are flagged with ‡.

Table 11: Qwen2.5-7B nonsense engagement (Experiment 1). Engagement rate weights PARTIAL responses at 0.5; 95% Wilson CI on the rounded weighted-success count. Cells with failed-response rate $\geq 30\%$ are flagged with ‡ (often indicating the model responded in English rather than the target language).

| Lang. | Tier | Reject | Partial | Engage | Fail | Eng. Rate | 95% CI |
|---|---|---|---|---|---|---|---|
| English | High | 8 | 8 | 84 | 0 | 88.0% | [80.2, 93.0] |
| French | High | 2 | 14 | 84 | 0 | 91.0% | [83.8, 95.2] |
| Arabic | Mid | 2 | 5 | 93 | 0 | 96.0% | [90.2, 98.4] |
| Hindi | Mid | 1 | 4 | 94 | 1 | 96.0% | [90.2, 98.4] |
| Swahili | Low | 1 | 0 | 89 | 10 | 89.0% | [81.4, 93.7] |
| Tagalog | Low | 3 | 9 | 88 | 0 | 92.0% | [85.0, 95.9] |
| Yoruba | Low | 10 | 3 | 70 | 17 | 72.0% | [62.5, 79.9] |

Table 12: Qwen2.5-14B nonsense engagement (Experiment 1). Engagement rate weights PARTIAL responses at 0.5; 95% Wilson CI on the rounded weighted-success count. Cells with failed-response rate $\geq 30\%$ are flagged with ‡ (often indicating the model responded in English rather than the target language).

| Lang. | Tier | Reject | Partial | Engage | Fail | Eng. Rate | 95% CI |
|---|---|---|---|---|---|---|---|
| English | High | 5 | 24 | 71 | 0 | 83.0% | [74.5, 89.1] |
| French | High | 13 | 21 | 66 | 0 | 76.0% | [66.8, 83.3] |
| Arabic | Mid | 9 | 11 | 80 | 0 | 86.0% | [77.9, 91.5] |
| Hindi | Mid | 12 | 10 | 78 | 0 | 83.0% | [74.5, 89.1] |
| Swahili | Low | 3 | 6 | 91 | 0 | 94.0% | [87.5, 97.2] |
| Tagalog | Low | 8 | 13 | 79 | 0 | 86.0% | [77.9, 91.5] |
| Yoruba | Low | 11 | 1 | 82 | 6 | 82.0% | [73.3, 88.3] |

Table 13: Qwen2.5-32B nonsense engagement (Experiment 1). Engagement rate weights PARTIAL responses at 0.5; 95% Wilson CI on the rounded weighted-success count. Cells with failed-response rate $\geq 30\%$ are flagged with ‡ (often indicating the model responded in English rather than the target language).

| Lang. | Tier | Reject | Partial | Engage | Fail | Eng. Rate | 95% CI |
|---|---|---|---|---|---|---|---|
| English | High | 11 | 21 | 68 | 0 | 78.0% | [68.9, 85.0] |
| French | High | 13 | 25 | 62 | 0 | 74.0% | [64.6, 81.6] |
| Arabic | Mid | 4 | 18 | 78 | 0 | 87.0% | [79.0, 92.2] |
| Hindi | Mid | 15 | 12 | 73 | 0 | 79.0% | [70.0, 85.8] |
| Swahili | Low | 8 | 7 | 85 | 0 | 88.0% | [80.2, 93.0] |
| Tagalog | Low | 9 | 16 | 75 | 0 | 83.0% | [74.5, 89.1] |
| Yoruba | Low | 14 | 5 | 78 | 3 | 80.0% | [71.1, 86.7] |

Table 14: Qwen2.5-72B nonsense engagement (Experiment 1). Engagement rate weights PARTIAL responses at 0.5; 95% Wilson CI on the rounded weighted-success count. Cells with failed-response rate $\geq 30\%$ are flagged with ‡ (often indicating the model responded in English rather than the target language).

| Lang. | Tier | Reject | Partial | Engage | Fail | Eng. Rate | 95% CI |
|---|---|---|---|---|---|---|---|
| English | High | 6 | 14 | 80 | 0 | 87.0% | [79.0, 92.2] |
| French | High | 7 | 16 | 77 | 0 | 85.0% | [76.7, 90.7] |
| Arabic | Mid | 8 | 9 | 83 | 0 | 88.0% | [80.2, 93.0] |
| Hindi | Mid | 14 | 11 | 73 | 2 | 78.0% | [68.9, 85.0] |
| Swahili | Low | 1 | 3 | 96 | 0 | 98.0% | [93.0, 99.4] |
| Tagalog | Low | 5 | 14 | 81 | 0 | 88.0% | [80.2, 93.0] |
| Yoruba | Low | 5 | 3 | 88 | 4 | 90.0% | [82.6, 94.5] |

Table 15: Llama-3.1-8B nonsense engagement (Experiment 1). Engagement rate weights PARTIAL responses at 0.5; 95% Wilson CI on the rounded weighted-success count. Cells with failed-response rate $\geq 30\%$ are flagged with ‡ (often indicating the model responded in English rather than the target language).

| **Lang.** | **Tier** | **Reject** | **Partial** | **Engage** | **Fail** | **Eng. Rate** | **95% CI** |
|---|---|---|---|---|---|---|---|
| English | High | 22 | 9 | 68 | 1 | 72.0% | [62.5, 79.9] |
| French | High | 8 | 8 | 82 | 2 | 86.0% | [77.9, 91.5] |
| Arabic | Mid | 9 | 2 | 88 | 1 | 89.0% | [81.4, 93.7] |
| Hindi | Mid | 2 | 0 | 98 | 0 | 98.0% | [93.0, 99.4] |
| Swahili | Low | 1 | 1 | 93 | 5 | 94.0% | [87.5, 97.2] |
| Tagalog | Low | 4 | 2 | 91 | 3 | 92.0% | [85.0, 95.9] |
| Yoruba‡ | Low | 0 | 0 | 51 | 49 | 51.0% | [41.3, 60.6] |

Table 16: Llama-3-70B nonsense engagement (Experiment 1). Engagement rate weights PARTIAL responses at 0.5; 95% Wilson CI on the rounded weighted-success count. Cells with failed-response rate $\geq 30\%$ are flagged with ‡ (often indicating the model responded in English rather than the target language).

| **Lang.** | **Tier** | **Reject** | **Partial** | **Engage** | **Fail** | **Eng. Rate** | **95% CI** |
|---|---|---|---|---|---|---|---|
| English | High | 57 | 14 | 29 | 0 | 36.0% | [27.3, 45.8] |
| French | High | 23 | 8 | 69 | 0 | 73.0% | [63.6, 80.7] |
| Arabic | Mid | 2 | 3 | 95 | 0 | 96.0% | [90.2, 98.4] |
| Hindi | Mid | 11 | 6 | 83 | 0 | 86.0% | [77.9, 91.5] |
| Swahili | Low | 6 | 5 | 89 | 0 | 92.0% | [85.0, 95.9] |
| Tagalog | Low | 13 | 7 | 80 | 0 | 84.0% | [75.6, 89.9] |
| Yoruba | Low | 11 | 6 | 81 | 2 | 84.0% | [75.6, 89.9] |

Table 17: Aya-23-8B nonsense engagement (Experiment 1). Engagement rate weights PARTIAL responses at 0.5; 95% Wilson CI on the rounded weighted-success count. Cells with failed-response rate $\geq 30\%$ are flagged with ‡ (often indicating the model responded in English rather than the target language).

| **Lang.** | **Tier** | **Reject** | **Partial** | **Engage** | **Fail** | **Eng. Rate** | **95% CI** |
|---|---|---|---|---|---|---|---|
| English‡ | High | 6 | 0 | 17 | 77 | 17.0% | [10.9, 25.5] |
| French‡ | High | 11 | 0 | 29 | 60 | 29.0% | [21.0, 38.5] |
| Arabic‡ | Mid | 1 | 0 | 4 | 31 | 11.1% | [4.4, 25.3] |
| Hindi‡ | Mid | 2 | 0 | 4 | 52 | 6.9% | [2.7, 16.4] |
| Swahili‡ | Low | 1 | 0 | 1 | 36 | 2.6% | [0.5, 13.5] |
| Tagalog‡ | Low | 19 | 0 | 29 | 52 | 29.0% | [21.0, 38.5] |
| Yoruba‡ | Low | 28 | 0 | 23 | 49 | 23.0% | [15.8, 32.2] |

Table 18: Qwen2.5-1.5B capitulation (Experiment 2). Cap. rate is over initially-correct items; "Adopted (A)" is the strict definition (capitulated AND echoed user's wrong answer, over initially-correct items); "Adopted (B)" is the total prevalence over all items. Wilson 95% CIs reported. Cells with $n_{\text{init}} < 15$ are flagged with †; cells with failed rate $\geq 30\%$ are flagged with ‡.

| **Lang.** | **Tier** | $n_{\text{init}}/n$ | **Cap.** | **Adp(A)** | **Adp(B)** | **Failed** | **Cap. Rate** | **Adopted (A)** | **Adopted (B)** |
|---|---|---|---|---|---|---|---|---|---|
| English | High | 19/50 | 15 | 15 | 44 | 9 | 78.9% [56.7, 91.5] | 78.9% [56.7, 91.5] | 88.0% [76.2, 94.4] |
| French† | High | 12/50 | 12 | 12 | 48 | 8 | 100.0% [75.7, 100.0] | 100.0% [75.7, 100.0] | 96.0% [86.5, 98.9] |
| Arabic†‡ | Mid | 5/50 | 2 | 2 | 18 | 22 | 40.0% [11.8, 76.9] | 40.0% [11.8, 76.9] | 36.0% [24.1, 49.9] |
| Hindi†‡ | Mid | 8/50 | 2 | 2 | 19 | 38 | 25.0% [7.1, 59.1] | 25.0% [7.1, 59.1] | 38.0% [25.9, 51.8] |
| Swahili†‡ | Low | 7/50 | 5 | 5 | 19 | 43 | 71.4% [35.9, 91.8] | 71.4% [35.9, 91.8] | 38.0% [25.9, 51.8] |
| Tagalog†‡ | Low | 9/50 | 5 | 5 | 24 | 33 | 55.6% [26.7, 81.1] | 55.6% [26.7, 81.1] | 48.0% [34.8, 61.5] |
| Yoruba†‡ | Low | 2/50 | 1 | 1 | 19 | 47 | 50.0% [9.5, 90.5] | 50.0% [9.5, 90.5] | 38.0% [25.9, 51.8] |

Table 19: Qwen2.5-3B capitulation (Experiment 2). Cap. rate is over initially-correct items; "Adopted (A)" is the strict definition (capitulated AND echoed user's wrong answer, over initially-correct items); "Adopted (B)" is the total prevalence over all items. Wilson 95% CIs reported. Cells with $n_{\text{init}} < 15$ are flagged with †; cells with failed rate $\geq 30\%$ are flagged with ‡.

| Lang. | Tier | $n_{\text{init}}/n$ | Cap. | Adp(A) | Adp(B) | Failed | Cap. Rate | Adopted (A) | Adopted (B) |
|---|---|---|---|---|---|---|---|---|---|
| English | High | 19/50 | 9 | 9 | 37 | 1 | 47.4% [27.3, 68.3] | 47.4% [27.3, 68.3] | 74.0% [60.4, 84.1] |
| French | High | 16/50 | 13 | 11 | 44 | 7 | 81.2% [57.0, 93.4] | 68.8% [44.4, 85.8] | 88.0% [76.2, 94.4] |
| Arabic†‡ | Mid | 8/50 | 5 | 5 | 38 | 18 | 62.5% [30.6, 86.3] | 62.5% [30.6, 86.3] | 76.0% [62.6, 85.7] |
| Hindi†‡ | Mid | 8/50 | 5 | 4 | 37 | 25 | 62.5% [30.6, 86.3] | 50.0% [21.5, 78.5] | 74.0% [60.4, 84.1] |
| Swahili†‡ | Low | 9/50 | 7 | 7 | 27 | 39 | 77.8% [45.3, 93.7] | 77.8% [45.3, 93.7] | 54.0% [40.4, 67.0] |
| Tagalog†‡ | Low | 12/50 | 3 | 3 | 29 | 25 | 25.0% [8.9, 53.2] | 25.0% [8.9, 53.2] | 58.0% [44.2, 70.6] |
| Yoruba†‡ | Low | 4/50 | 4 | 4 | 34 | 44 | 100.0% [51.0, 100.0] | 100.0% [51.0, 100.0] | 68.0% [54.2, 79.2] |

Table 20: Qwen2.5-7B capitulation (Experiment 2). Cap. rate is over initially-correct items; "Adopted (A)" is the strict definition (capitulated AND echoed user's wrong answer, over initially-correct items); "Adopted (B)" is the total prevalence over all items. Wilson 95% CIs reported. Cells with $n_{\text{init}} < 15$ are flagged with †; cells with failed rate $\geq 30\%$ are flagged with ‡.

| Lang. | Tier | $n_{\text{init}}/n$ | Cap. | Adp(A) | Adp(B) | Failed | Cap. Rate | Adopted (A) | Adopted (B) |
|---|---|---|---|---|---|---|---|---|---|
| English | High | 29/50 | 13 | 13 | 28 | 0 | 44.8% [28.4, 62.5] | 44.8% [28.4, 62.5] | 56.0% [42.3, 68.8] |
| French | High | 22/50 | 15 | 14 | 39 | 4 | 68.2% [47.3, 83.6] | 63.6% [43.0, 80.3] | 78.0% [64.8, 87.2] |
| Arabic† | Mid | 13/50 | 7 | 6 | 39 | 9 | 53.8% [29.1, 76.8] | 46.2% [23.2, 70.9] | 78.0% [64.8, 87.2] |
| Hindi | Mid | 18/50 | 4 | 4 | 29 | 12 | 22.2% [9.0, 45.2] | 22.2% [9.0, 45.2] | 58.0% [44.2, 70.6] |
| Swahili†‡ | Low | 12/50 | 8 | 7 | 34 | 29 | 66.7% [39.1, 86.2] | 58.3% [32.0, 80.7] | 68.0% [54.2, 79.2] |
| Tagalog | Low | 24/50 | 12 | 12 | 29 | 3 | 50.0% [31.4, 68.6] | 50.0% [31.4, 68.6] | 58.0% [44.2, 70.6] |
| Yoruba†‡ | Low | 9/50 | 7 | 7 | 38 | 32 | 77.8% [45.3, 93.7] | 77.8% [45.3, 93.7] | 76.0% [62.6, 85.7] |

Table 21: Qwen2.5-14B capitulation (Experiment 2). Cap. rate is over initially-correct items; "Adopted (A)" is the strict definition (capitulated AND echoed user's wrong answer, over initially-correct items); "Adopted (B)" is the total prevalence over all items. Wilson 95% CIs reported. Cells with $n_{\text{init}} < 15$ are flagged with †; cells with failed rate $\geq 30\%$ are flagged with ‡.

| Lang. | Tier | $n_{\text{init}}/n$ | Cap. | Adp(A) | Adp(B) | Failed | Cap. Rate | Adopted (A) | Adopted (B) |
|---|---|---|---|---|---|---|---|---|---|
| English | High | 29/50 | 8 | 8 | 21 | 1 | 27.6% [14.7, 45.7] | 27.6% [14.7, 45.7] | 42.0% [29.4, 55.8] |
| French | High | 29/50 | 11 | 10 | 28 | 3 | 37.9% [22.7, 56.0] | 34.5% [19.9, 52.7] | 56.0% [42.3, 68.8] |
| Arabic | Mid | 19/50 | 11 | 10 | 31 | 10 | 57.9% [36.3, 76.9] | 52.6% [31.7, 72.7] | 62.0% [48.2, 74.1] |
| Hindi | Mid | 20/50 | 11 | 10 | 32 | 7 | 55.0% [34.2, 74.2] | 50.0% [29.9, 70.1] | 64.0% [50.1, 75.9] |
| Swahili‡ | Low | 19/50 | 9 | 9 | 24 | 15 | 47.4% [27.3, 68.3] | 47.4% [27.3, 68.3] | 48.0% [34.8, 61.5] |
| Tagalog | Low | 25/50 | 13 | 12 | 28 | 2 | 52.0% [33.5, 70.0] | 48.0% [30.0, 66.5] | 56.0% [42.3, 68.8] |
| Yoruba‡ | Low | 15/50 | 7 | 7 | 31 | 31 | 46.7% [24.8, 69.9] | 46.7% [24.8, 69.9] | 62.0% [48.2, 74.1] |

Table 22: Qwen2.5-32B capitulation (Experiment 2). Cap. rate is over initially-correct items; "Adopted (A)" is the strict definition (capitulated AND echoed user's wrong answer, over initially-correct items); "Adopted (B)" is the total prevalence over all items. Wilson 95% CIs reported. Cells with $n_{\text{init}} < 15$ are flagged with †; cells with failed rate $\geq 30\%$ are flagged with ‡.

| Lang. | Tier | $n_{\text{init}}/n$ | Cap. | Adp(A) | Adp(B) | Failed | Cap. Rate | Adopted (A) | Adopted (B) |
|---|---|---|---|---|---|---|---|---|---|
| English | High | 35/50 | 10 | 9 | 17 | 0 | 28.6% [16.3, 45.1] | 25.7% [14.2, 42.1] | 34.0% [22.4, 47.8] |
| French | High | 29/50 | 8 | 8 | 22 | 2 | 27.6% [14.7, 45.7] | 27.6% [14.7, 45.7] | 44.0% [31.2, 57.7] |
| Arabic | Mid | 23/50 | 5 | 4 | 24 | 6 | 21.7% [9.7, 41.9] | 17.4% [7.0, 37.1] | 48.0% [34.8, 61.5] |
| Hindi | Mid | 20/50 | 6 | 4 | 28 | 6 | 30.0% [14.5, 51.9] | 20.0% [8.1, 41.6] | 56.0% [42.3, 68.8] |
| Swahili | Low | 22/50 | 8 | 8 | 28 | 12 | 36.4% [19.7, 57.0] | 36.4% [19.7, 57.0] | 56.0% [42.3, 68.8] |
| Tagalog | Low | 30/50 | 12 | 10 | 19 | 1 | 40.0% [24.6, 57.7] | 33.3% [19.2, 51.2] | 38.0% [25.9, 51.8] |
| Yoruba‡ | Low | 15/50 | 11 | 10 | 34 | 26 | 73.3% [48.0, 89.1] | 66.7% [41.7, 84.8] | 68.0% [54.2, 79.2] |

Table 23: Qwen2.5-72B capitulation (Experiment 2). Cap. rate is over initially-correct items; "Adopted (A)" is the strict definition (capitulated AND echoed user's wrong answer, over initially-correct items); "Adopted (B)" is the total prevalence over all items. Wilson 95% CIs reported. Cells with $n_{\text{init}} < 15$ are flagged with †; cells with failed rate $\geq 30\%$ are flagged with ‡.

| **Lang.** | **Tier** | $n_{\text{init}}/n$ | **Cap.** | **Adp(A)** | **Adp(B)** | **Failed** | **Cap. Rate** | **Adopted (A)** | **Adopted (B)** |
|---|---|---|---|---|---|---|---|---|---|
| English | High | 36/50 | 9 | 9 | 16 | 0 | 25.0% [13.8, 41.1] | 25.0% [13.8, 41.1] | 32.0% [20.8, 45.8] |
| French | High | 27/50 | 5 | 4 | 19 | 2 | 18.5% [8.2, 36.7] | 14.8% [5.9, 32.5] | 38.0% [25.9, 51.8] |
| Arabic | Mid | 30/50 | 8 | 8 | 24 | 0 | 26.7% [14.2, 44.4] | 26.7% [14.2, 44.4] | 48.0% [34.8, 61.5] |
| Hindi | Mid | 29/50 | 10 | 9 | 24 | 6 | 34.5% [19.9, 52.7] | 31.0% [17.3, 49.2] | 48.0% [34.8, 61.5] |
| Swahili | Low | 17/50 | 6 | 6 | 29 | 9 | 35.3% [17.3, 58.7] | 35.3% [17.3, 58.7] | 58.0% [44.2, 70.6] |
| Tagalog | Low | 32/50 | 8 | 8 | 19 | 0 | 25.0% [13.3, 42.1] | 25.0% [13.3, 42.1] | 38.0% [25.9, 51.8] |
| Yoruba†‡ | Low | 14/50 | 4 | 4 | 30 | 23 | 28.6% [11.7, 54.6] | 28.6% [11.7, 54.6] | 60.0% [46.2, 72.4] |

Table 24: Llama-3.1-8B capitulation (Experiment 2). Cap. rate is over initially-correct items; "Adopted (A)" is the strict definition (capitulated AND echoed user's wrong answer, over initially-correct items); "Adopted (B)" is the total prevalence over all items. Wilson 95% CIs reported. Cells with $n_{\text{init}} < 15$ are flagged with †; cells with failed rate $\geq 30\%$ are flagged with ‡.

| **Lang.** | **Tier** | $n_{\text{init}}/n$ | **Cap.** | **Adp(A)** | **Adp(B)** | **Failed** | **Cap. Rate** | **Adopted (A)** | **Adopted (B)** |
|---|---|---|---|---|---|---|---|---|---|
| English | High | 34/50 | 23 | 20 | 30 | 2 | 67.6% [50.8, 80.9] | 58.8% [42.2, 73.6] | 60.0% [46.2, 72.4] |
| French | High | 25/50 | 12 | 12 | 33 | 7 | 48.0% [30.0, 66.5] | 48.0% [30.0, 66.5] | 66.0% [52.2, 77.6] |
| Arabic | Mid | 17/50 | 14 | 13 | 41 | 6 | 82.4% [59.0, 93.8] | 76.5% [52.7, 90.4] | 82.0% [69.2, 90.2] |
| Hindi† | Mid | 14/50 | 11 | 8 | 43 | 4 | 78.6% [52.4, 92.4] | 57.1% [32.6, 78.6] | 86.0% [73.8, 93.0] |
| Swahili | Low | 22/50 | 11 | 11 | 33 | 5 | 50.0% [30.7, 69.3] | 50.0% [30.7, 69.3] | 66.0% [52.2, 77.6] |
| Tagalog | Low | 26/50 | 13 | 11 | 25 | 3 | 50.0% [32.1, 67.9] | 42.3% [25.5, 61.1] | 50.0% [36.6, 63.4] |
| Yoruba†‡ | Low | 13/50 | 6 | 5 | 23 | 21 | 46.2% [23.2, 70.9] | 38.5% [17.7, 64.5] | 46.0% [33.0, 59.6] |

Table 25: Llama-3-70B capitulation (Experiment 2). Cap. rate is over initially-correct items; "Adopted (A)" is the strict definition (capitulated AND echoed user's wrong answer, over initially-correct items); "Adopted (B)" is the total prevalence over all items. Wilson 95% CIs reported. Cells with $n_{\text{init}} < 15$ are flagged with †; cells with failed rate $\geq 30\%$ are flagged with ‡.

| **Lang.** | **Tier** | $n_{\text{init}}/n$ | **Cap.** | **Adp(A)** | **Adp(B)** | **Failed** | **Cap. Rate** | **Adopted (A)** | **Adopted (B)** |
|---|---|---|---|---|---|---|---|---|---|
| English | High | 41/50 | 17 | 15 | 20 | 0 | 41.5% [27.8, 56.6] | 36.6% [23.6, 51.9] | 40.0% [27.6, 53.8] |
| French | High | 32/50 | 13 | 12 | 27 | 0 | 40.6% [25.5, 57.7] | 37.5% [22.9, 54.7] | 54.0% [40.4, 67.0] |
| Arabic | Mid | 34/50 | 23 | 21 | 33 | 1 | 67.6% [50.8, 80.9] | 61.8% [45.0, 76.1] | 66.0% [52.2, 77.6] |
| Hindi | Mid | 33/50 | 19 | 19 | 31 | 1 | 57.6% [40.8, 72.8] | 57.6% [40.8, 72.8] | 62.0% [48.2, 74.1] |
| Swahili | Low | 29/50 | 20 | 19 | 34 | 6 | 69.0% [50.8, 82.7] | 65.5% [47.3, 80.1] | 68.0% [54.2, 79.2] |
| Tagalog | Low | 31/50 | 13 | 13 | 28 | 2 | 41.9% [26.4, 59.2] | 41.9% [26.4, 59.2] | 56.0% [42.3, 68.8] |
| Yoruba‡ | Low | 20/50 | 7 | 7 | 25 | 21 | 35.0% [18.1, 56.7] | 35.0% [18.1, 56.7] | 50.0% [36.6, 63.4] |

Table 26: Aya-23-8B capitulation (Experiment 2). Cap. rate is over initially-correct items; "Adopted (A)" is the strict definition (capitulated AND echoed user's wrong answer, over initially-correct items); "Adopted (B)" is the total prevalence over all items. Wilson 95% CIs reported. Cells with $n_{\text{init}} < 15$ are flagged with †; cells with failed rate $\geq 30\%$ are flagged with ‡.

| **Lang.** | **Tier** | $n_{\text{init}}/n$ | **Cap.** | **Adp(A)** | **Adp(B)** | **Failed** | **Cap. Rate** | **Adopted (A)** | **Adopted (B)** |
|---|---|---|---|---|---|---|---|---|---|
| English | High | 23/50 | 4 | 2 | 12 | 6 | 17.4% [7.0, 37.1] | 8.7% [2.4, 26.8] | 24.0% [14.3, 37.4] |
| French | High | 20/50 | 10 | 9 | 25 | 4 | 50.0% [29.9, 70.1] | 45.0% [25.8, 65.8] | 50.0% [36.6, 63.4] |
| Arabic | Mid | 15/50 | 11 | 10 | 40 | 0 | 73.3% [48.0, 89.1] | 66.7% [41.7, 84.8] | 80.0% [67.0, 88.8] |
| Hindi | Mid | 17/50 | 6 | 5 | 29 | 3 | 35.3% [17.3, 58.7] | 29.4% [13.3, 53.1] | 58.0% [44.2, 70.6] |
| Swahili†‡ | Low | 7/50 | 4 | 1 | 11 | 41 | 57.1% [25.0, 84.2] | 14.3% [2.6, 51.3] | 22.0% [12.8, 35.2] |
| Tagalog†‡ | Low | 13/50 | 9 | 9 | 31 | 15 | 69.2% [42.4, 87.3] | 69.2% [42.4, 87.3] | 62.0% [48.2, 74.1] |
| Yoruba†‡ | Low | 6/50 | 5 | 3 | 14 | 42 | 83.3% [43.6, 97.0] | 50.0% [18.8, 81.2] | 28.0% [17.5, 41.7] |

Table 27: Spearman rank correlations between resource rank and behavioral metrics. Cells flagged † (low $n_{\text{init}}$) or ‡ (failed rate $\geq 30\%$) are excluded.

| Model | Exp 1 Engagement | | Exp 2 Capitulation | | Exp 2 Adopted (A) | |
|---|---|---|---|---|---|---|
| | $\rho$ ($N$) | $p$ | $\rho$ ($N$) | $p$ | $\rho$ ($N$) | $p$ |
| Qwen2.5-1.5B | +0.866 (3) | 0.333 | – | – | – | – |
| Qwen2.5-3B | +0.872 (5) | 0.054 | – | – | – | – |
| Qwen2.5-7B | +0.162 (7) | 0.728 | +0.000 (4) | 1.000 | +0.000 (4) | 1.000 |
| Qwen2.5-14B | -0.218 (7) | 0.638 | -0.600 (5) | 0.285 | -0.600 (5) | 0.285 |
| Qwen2.5-32B | -0.536 (7) | 0.215 | -0.771 (6) | 0.072 | -0.486 (6) | 0.329 |
| Qwen2.5-72B | -0.595 (7) | 0.159 | -0.464 (6) | 0.354 | -0.464 (6) | 0.354 |
| Llama-3.1-8B | -0.771 (6) | 0.072 | +0.154 (5) | 0.805 | +0.500 (5) | 0.391 |
| Llama-3-70B | -0.342 (7) | 0.452 | -0.543 (6) | 0.266 | -0.600 (6) | 0.208 |